

Mahir Patel

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Education

University of California - Riverside

MS in Computer Science, GPA: 4.0

Riverside, California

September 2025 - June 2026

University of California - Riverside

BS in Computer Science, GPA: 3.9

Riverside, California

September 2022 - April 2025

Experience

Co-Founder & Lead Backend Engineer @ EfficiencyMate

Remote

Full-Stack Development | Infrastructure & DevOps

December 2025 - Present

- Architected production backend with Python (FastAPI) and Google Gemini LLM, serving **50+** users with **25%** reported time savings and validated product-market fit
- Engineered AWS infrastructure (RDS, S3, VPC, App Runner) with database optimization and containerized deployment, reducing costs **40%** through ECS-to-App Runner migration
- Engineered full-stack architecture for horizontal scaling with database optimization, query indexing, and containerized deployment strategies supporting growing user base.

Graduate Researcher @UCR - Advisor: Dr. Yue Dong

Riverside, California

Deep Learning Engineer | NLP Researcher

June 2025 - Present

- Investigating mechanistic interpretability in LLMs using PyTorch and TransformerLens, developing custom analysis frameworks to decompose hidden state representations across model families
- Built evaluation tools to quantify reasoning circuits and model behavior attribution, conducting systematic benchmarking across diverse transformer architectures
- Collaborating on multi-faceted research including hallucination detection mechanisms and steering interventions to improve factuality in language models

Robotics Lab UCR

Riverside, California

Software Engineering | Embedded Systems

Sept 2023 - June 2025

- Implemented Raspberry Pi face recognition system improving accuracy by **15%** through hardware-software co-optimization
- Accelerated delivery of 3 robotic solutions by **20%** through cross-functional collaboration with research team
- Reduced system downtime **30%** by diagnosing and resolving **10+** critical hardware-software integration issues
- Maintained **98.7%** customer satisfaction delivering products **15%** faster using Agile methodology

Projects

LiREF: Steering for Llama-3.1-8B | Python, PyTorch, Transformers, LLM Steering

[github.com/steering-vector](https://github.com/mahirpatel/steering-vector)

- Designed a layer-wise LiREF steering framework to compute semantic vectors across 32 layers.
- Engineered a last-token steering mechanism via forward hooks, adding **<1%** runtime overhead.
- Achieved fine-grained causal control over model reasoning with ± 0.05 α -scaled hidden state shifts.
- Validated representational alignment on HotpotQA, showing accuracy gains ($\Delta_p \approx +2.3\%$).

High-Performance GPT-2 in C++ | C++, OpenMP, SIMD, CPU Optimization

[github.com/gpt2-c++](https://github.com/mahirpatel/gpt2-c++)

- Built GPT-2 from scratch in C++ for high-performance training.
- Hand-tuned `matmul_backward` kernel, achieving a **1.22x** overall training speedup.
- Optimized gradient computation using OpenMP, loop tiling, and SIMD.
- Reduced L1 dcache miss rate to **10.2%** via cache-aware tiling and builtin prefetch.

Technical Skills

Languages: Python, C, C++, Rust, TypeScript, SQL, Bash, Assembly

Developer Tools: Pytorch, FastAPI, LangGraph, LangChain, Transformers, HuggingFace, Pandas, NumPy, CUDA, React, NextJS

Technologies: AI, ML, Deep Learning, GPU programming, OS, Networks, AWS (RDS, S3, App Runner, VPC)